

we will give a detailed overview of the hardware and a list of necessary tools you will require for the Do-It-Yourself projects we have suggested. As a geek bonus, we also outline how you can create your own operating system on the Raspberry Pi. So brew up a pot of coffee and settle in for a fun and educational ride.

Hardware Components of the Pi

The Raspberry Pi and the newest Raspberry Pi 2 are fairly similar only in the top range of their models. For the original Raspberry Pi the Model B and Model B+ are at the top of the line, while there is only one of Model B for the Raspberry Pi 2

which has been released so far. There are only two points of difference between the two - the new and improved 900 MHz quad-core ARM Cortex-A7 CPU and the enhanced 1 GB of RAM. Apart for these two upgrades the



The Pi no bigger than a credit card in its form factor.

Raspberry Pi 2 is identical in every respect. Thanks to the new processor the microcomputer can now run advanced operating systems like Snappy Ubuntu Core and Microsoft Windows 10.

The key hardware components for the Raspberry Model B and B+ are the same. They come equipped with a microSD slot, 40 GPIO pins, four USB ports, micro USB port, 3.5 mm audio port, Ethernet port, CSI connection for camera interface, display interface, HDMI port and a VideoCore IV 3D graphics core. All these components come as a part of the System on a Chip design with the Broadcom BCM2835 and BCM2836 for Raspberry Pi and Pi 2 respectively.

First Time Booting

In earlier chapters we have already discussed how you can easily install an operating system like Raspbian or Pidora onto your Pi. Now we will learn the basic environment in which the Pi user interface exists and how best to make the most of its features.

After you have booted up the Pi to your operating system you will arrive at the Config tool by default. This is the central hub from which you can